

Relations and functions



1

x	y
2	2
4	4
6	3
7	5

2

- | | | | |
|----------------|----------------|----------------|----------------|
| a True | b False | c True | d True |
| e False | f False | g False | h False |
| i True | j False | k False | l False |

3

- | | |
|---|---|
| a Both a relation and a function | b Relation only |
| c Relation only | d Relation only |
| e Both a relation and a function | f Both a relation and a function |
| g Relation only | h Both a relation and a function |
| i Both a relation and a function | j Relation only |

4

- | | | |
|--------------------------|----------------------|--------------|
| a 16, 8, -15, -20 | b 14, 28, -28 | c Yes |
|--------------------------|----------------------|--------------|

5

- | | |
|---|---|
| a Both a function and a relation | b Relation |
| c Relation | d Both a function and a relation |
| e Relation | |

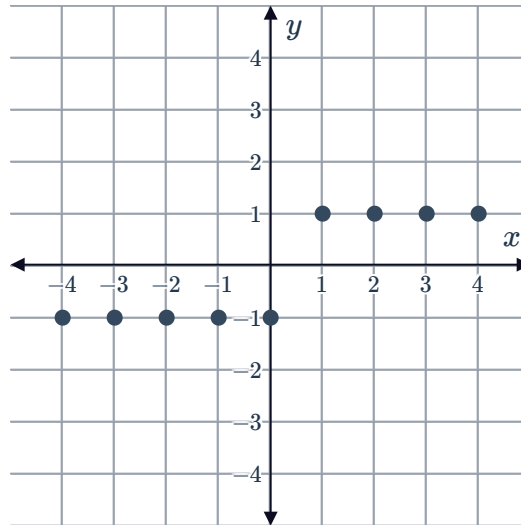
6



a

x	-4	-3	-2	-1	0	1	2	3	4
y	-1	-1	-1	-1	-1	1	1	1	1

b



c Yes

7

a Yes

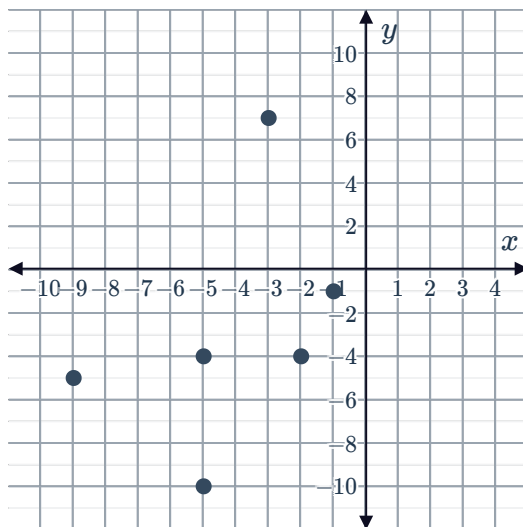
b Yes

c Yes

d No

8

a



b No

c Either $(-5, -10)$ or $(-5, -4)$

9



Function notation

10

a 46

b -189

c $\frac{33}{4}$

d 87

11

a $g(7) = -6$

b $f(x) = \frac{1}{4}x - 2$

12

a 2

b 0

c 1

13

a 1

b Yes

14

a 12

b -16

15

a 8

b -13

16

a $\frac{35}{4}$

b -7

17

a 1

b $-\frac{13}{5}$

18

a 68

b -4

19 $\frac{9}{2}$

20

a 0

b 3

c $\frac{91}{25}$



21

- a i -48 ii 15 iii -49

b $x = \pm 7.81$

22

a $a^2 + 8a$

b $b^2 + 8b$

23

a -4

b $-\frac{121}{32}$

24

a 0

b 2.67

c 80.99

25 -46

26

a 26

b 52

c 32

27

a $\sqrt{128}$

b 12

c $\sqrt{63}$

d $\sqrt{142}$

28

a 12

b $m^2 + 8$

29

a 39

b -25

c $a^3 - 8a + 7$

d $b^3 - 8b + 7$

e $(a + b)^3 - 8(a + b) + 7$

f No

30

a i $f(x) = 2 - \frac{1}{3}x$

ii 1

b i $f(x) = \frac{6}{5}x + \frac{7}{5}$

ii 5

c i $f(x) = 3 - x - 6x^2$

ii -54



31

a

i $f(x) = \frac{7-5x}{-3}$

ii 1

b

i $f(x) = 4x^2 + x + 5$

ii 23

32

a

$f(x) = \frac{1}{4}x - 2$

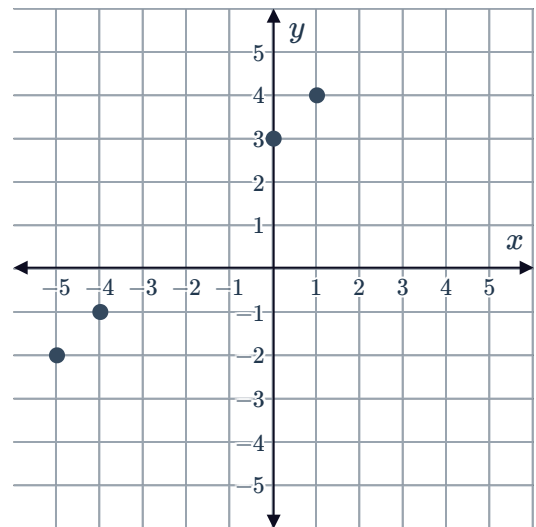
b 1

33

a

x	-5	-4	0	1
$f(x)$	-2	-1	3	4

b

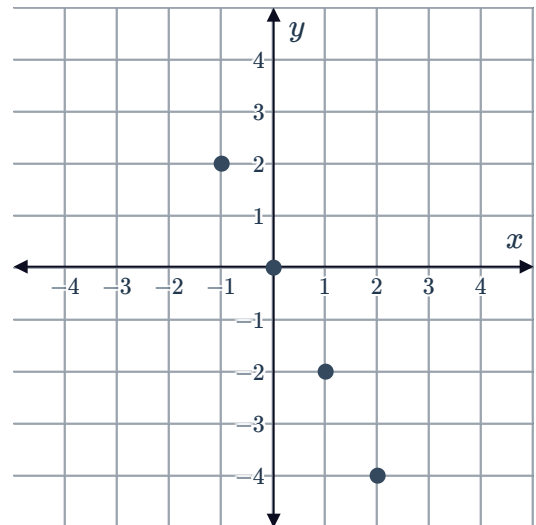


34

a

x	-1	0	1	2
$f(x)$	2	0	-2	-4

b



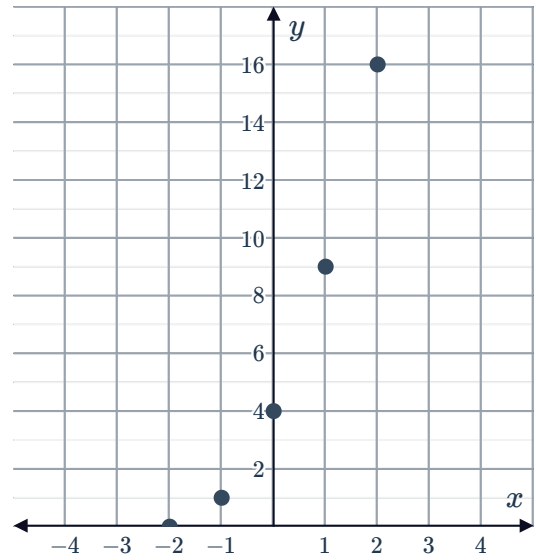
35

a

x	-2	-1	0	1	2
$f(x)$	0	1	4	9	16



b

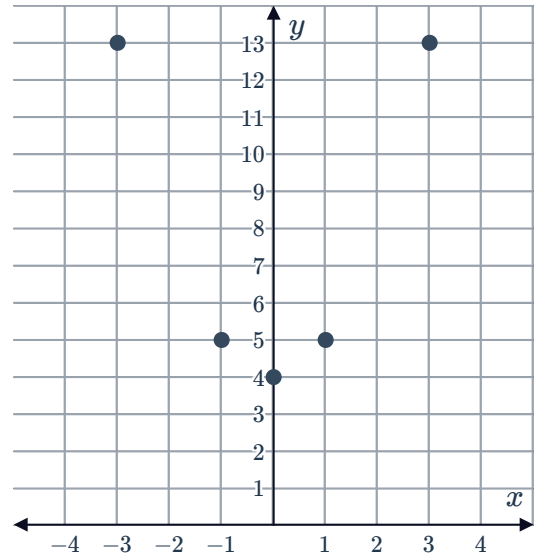


36

a

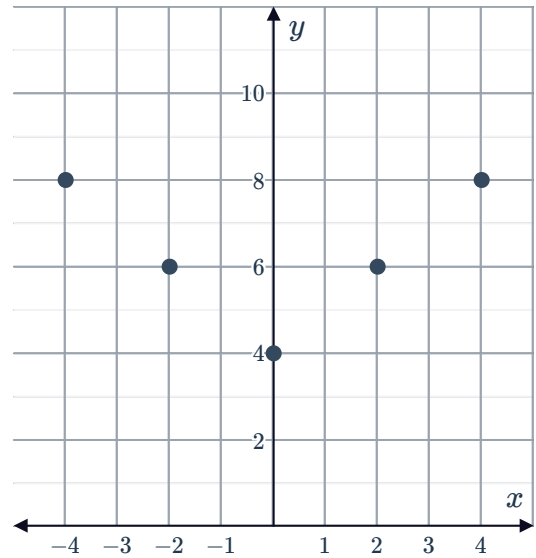
x	-3	-1	0	1	3
$f(x)$	13	5	4	5	13

b



37 a

x	-4	-2	0	2	4
$f(x)$	8	6	4	6	8



Applications

38 a

Call length (minutes)	International Call cost
1	\$3.50
2	\$6.40
3	\$9.30
4	\$12.20
5	\$15.10

b Yes

39 a

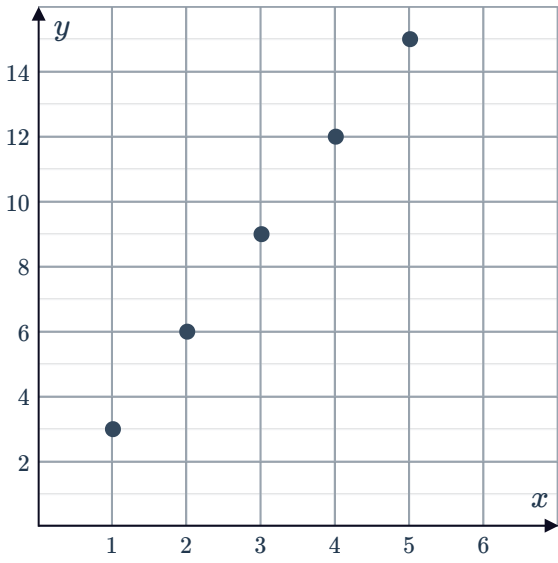
Number of T-shirts	Total Cost (dollars)
1	\$19
2	\$38
3	\$38
4	\$57
5	\$76

b Yes



40

a



b Yes